

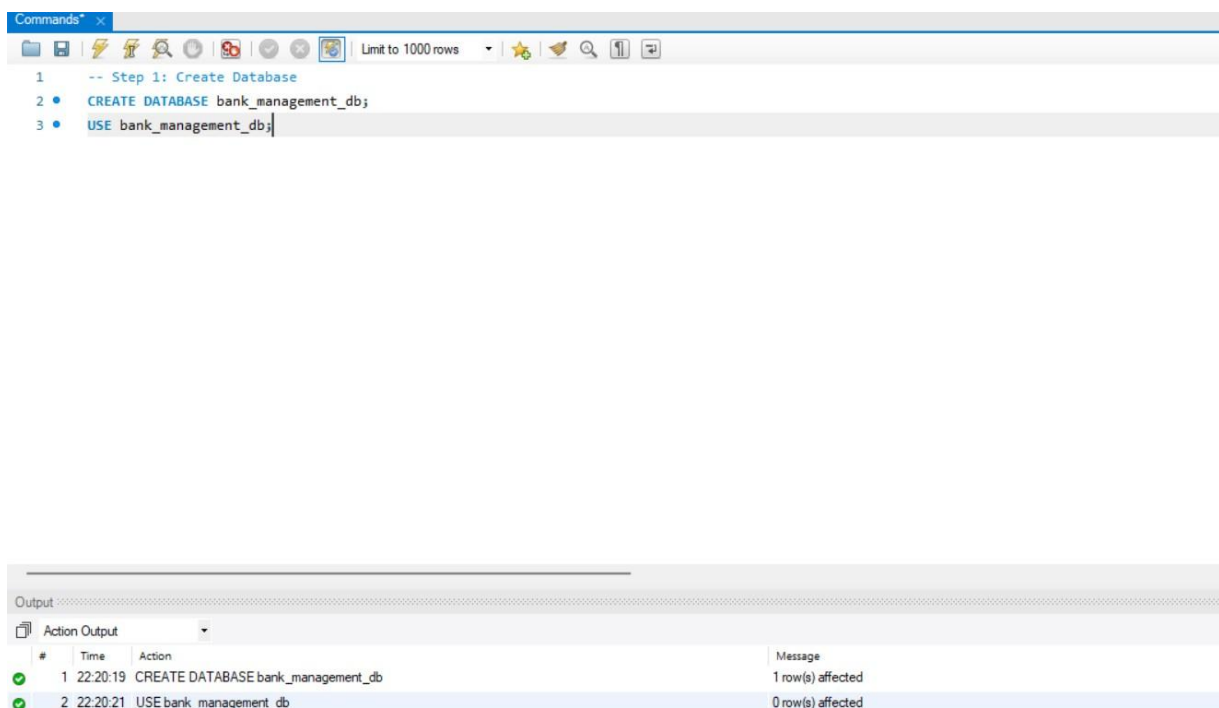
Commands

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Question :

Perform the following DDL, DML, DCL, and TCL operations on a Bank Management System database: create and alter a bank transaction table, add and modify columns, rename, truncate and drop tables; insert, update, delete, and retrieve transaction records using SELECT, WHERE, and ORDER BY; grant and revoke user privileges; and perform transaction control operations using COMMIT, SAVEPOINT, and ROLLBACK.

Step 1 — Create Database



The screenshot displays a database command window with a toolbar at the top. The command window contains the following SQL commands:

```
-- Step 1: Create Database
1  CREATE DATABASE bank_management_db;
2  USE bank_management_db;
```

Below the command window, the 'Output' tab is selected, showing a table with the execution results:

#	Time	Action	Message
1	22:20:19	CREATE DATABASE bank_management_db	1 row(s) affected
2	22:20:21	USE bank_management_db	0 row(s) affected

Step 2 — DDL Commands

```
Commands* x
Limit to 1000 rows

1  -- Step 1: Create Database
2  • CREATE DATABASE bank_management_db;
3  • USE bank_management_db;
4  -- Step 2: Create Bank Transactions Table
5  • CREATE TABLE bank_transactions(
6      txn_id INT PRIMARY KEY,
7      customer_name VARCHAR(50),
8      branch_name VARCHAR(50),
9      transaction_type VARCHAR(20),
10     amount DECIMAL(10,2),
11     transaction_date DATE
12 );
```

Output

Action Output

#	Time	Action	Message
✓ 1	22:20:19	CREATE DATABASE bank_management_db	1 row(s) affected
✓ 2	22:20:21	USE bank_management_db	0 row(s) affected
✓ 3	22:21:06	CREATE TABLE bank_transactions(txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_...	0 row(s) affected

```
Commands* x
Limit to 1000 rows

1  -- Step 1: Create Database
2  • CREATE DATABASE bank_management_db;
3  • USE bank_management_db;
4  -- Step 2: Create Bank Transactions Table
5  • CREATE TABLE bank_transactions(
6      txn_id INT PRIMARY KEY,
7      customer_name VARCHAR(50),
8      branch_name VARCHAR(50),
9      transaction_type VARCHAR(20),
10     amount DECIMAL(10,2),
11     transaction_date DATE
12 );
13 -- Step 3: Add Account Number Column
14 • ALTER TABLE bank_transactions
15 ADD account_no VARCHAR(20);
```

Output

Action Output

#	Time	Action	Message
✓ 1	22:20:19	CREATE DATABASE bank_management_db	1 row(s) affected
✓ 2	22:20:21	USE bank_management_db	0 row(s) affected
✓ 3	22:21:06	CREATE TABLE bank_transactions(txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_...	0 row(s) affected
✓ 4	22:21:32	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

```
Commands x
Limit to 1000 rows

1  -- Step 1: Create Database
2  • CREATE DATABASE bank_management_db;
3  • USE bank_management_db;
4  -- Step 2: Create Bank Transactions Table
5  • CREATE TABLE bank_transactions(
6      txn_id INT PRIMARY KEY,
7      customer_name VARCHAR(50),
8      branch_name VARCHAR(50),
9      transaction_type VARCHAR(20),
10     amount DECIMAL(10,2),
11     transaction_date DATE
12 );
13 -- Step 3: Add Account Number Column
14 • ALTER TABLE bank_transactions
15   ADD account_no VARCHAR(20);
16 -- Step 4: Modify Customer Name Column
17 • ALTER TABLE bank_transactions
18   MODIFY customer_name VARCHAR(100);
```

Output

Action Output

#	Time	Action	Message
✓ 1	22:20:19	CREATE DATABASE bank_management_db	1 row(s) affected
✓ 2	22:20:21	USE bank_management_db	0 row(s) affected
✓ 3	22:21:06	CREATE TABLE bank_transactions(txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_...	0 row(s) affected
✓ 4	22:21:32	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 5	22:22:07	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

```
Commands x
Limit to 1000 rows

2  • CREATE DATABASE bank_management_db;
3  • USE bank_management_db;
4  -- Step 2: Create Bank Transactions Table
5  • CREATE TABLE bank_transactions(
6      txn_id INT PRIMARY KEY,
7      customer_name VARCHAR(50),
8      branch_name VARCHAR(50),
9      transaction_type VARCHAR(20),
10     amount DECIMAL(10,2),
11     transaction_date DATE
12 );
13 -- Step 3: Add Account Number Column
14 • ALTER TABLE bank_transactions
15   ADD account_no VARCHAR(20);
16 -- Step 4: Modify Customer Name Column
17 • ALTER TABLE bank_transactions
18   MODIFY customer_name VARCHAR(100);
19 -- Step 5: Rename Transaction Table
20 • RENAME TABLE bank_transactions
21   TO customer_transactions;
```

Output

Action Output

#	Time	Action	Message
✓ 1	22:20:19	CREATE DATABASE bank_management_db	1 row(s) affected
✓ 2	22:20:21	USE bank_management_db	0 row(s) affected
✓ 3	22:21:06	CREATE TABLE bank_transactions(txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_...	0 row(s) affected
✓ 4	22:21:32	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 5	22:22:07	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 6	22:22:54	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected

Commands

```

4  -- Step 2: Create Bank Transactions Table
5  • CREATE TABLE bank_transactions(
6      txn_id INT PRIMARY KEY,
7      customer_name VARCHAR(50),
8      branch_name VARCHAR(50),
9      transaction_type VARCHAR(20),
10     amount DECIMAL(10,2),
11     transaction_date DATE
12 );
13  -- Step 3: Add Account Number Column
14  • ALTER TABLE bank_transactions
15     ADD account_no VARCHAR(20);
16  -- Step 4: Modify Customer Name Column
17  • ALTER TABLE bank_transactions
18     MODIFY customer_name VARCHAR(100);
19  -- Step 5: Rename Transaction Table
20  • RENAME TABLE bank_transactions
21     TO customer_transactions;
22  -- Step 6: Truncate Transaction Table
23  • TRUNCATE TABLE customer_transactions;

```

Output

Action Output

#	Time	Action	Message
✓ 2	22:20:21	USE bank_management_db	0 row(s) affected
✓ 3	22:21:06	CREATE TABLE bank_transactions(txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branc...	0 row(s) affected
✓ 4	22:21:32	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 5	22:22:07	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 6	22:22:54	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected
✓ 7	22:23:35	TRUNCATE TABLE customer_transactions	0 row(s) affected

Commands

```

9      transaction_type VARCHAR(20),
10     amount DECIMAL(10,2),
11     transaction_date DATE
12 );
13  -- Step 3: Add Account Number Column
14  • ALTER TABLE bank_transactions
15     ADD account_no VARCHAR(20);
16  -- Step 4: Modify Customer Name Column
17  • ALTER TABLE bank_transactions
18     MODIFY customer_name VARCHAR(100);
19  -- Step 5: Rename Transaction Table
20  • RENAME TABLE bank_transactions
21     TO customer_transactions;
22  -- Step 6: Truncate Transaction Table
23  • TRUNCATE TABLE customer_transactions;
24  -- Step 7: Create Backup Table
25  • CREATE TABLE bank_backup(
26     txn_id INT,
27     customer_name VARCHAR(100)
28 );

```

Output

Action Output

#	Time	Action	Message
✓ 3	22:21:06	CREATE TABLE bank_transactions(txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branc...	0 row(s) affected
✓ 4	22:21:32	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 5	22:22:07	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 6	22:22:54	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected
✓ 7	22:23:35	TRUNCATE TABLE customer_transactions	0 row(s) affected
✓ 8	22:24:10	CREATE TABLE bank_backup(txn_id INT, customer_name VARCHAR(100))	0 row(s) affected

Commands

```

11     transaction_date DATE
12 );
13 -- Step 3: Add Account Number Column
14 • ALTER TABLE bank_transactions
15   ADD account_no VARCHAR(20);
16 -- Step 4: Modify Customer Name Column
17 • ALTER TABLE bank_transactions
18   MODIFY customer_name VARCHAR(100);
19 -- Step 5: Rename Transaction Table
20 • RENAME TABLE bank_transactions
21   TO customer_transactions;
22 -- Step 6: Truncate Transaction Table
23 • TRUNCATE TABLE customer_transactions;
24 -- Step 7: Create Backup Table
25 • CREATE TABLE bank_backup(
26     txn_id INT,
27     customer_name VARCHAR(100)
28 );
29 -- Step 8: Drop Backup Table
30 • DROP TABLE bank_backup;

```

Output

Action Output

#	Time	Action	Message
✓ 4	22:21:32	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 5	22:22:07	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 6	22:22:54	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected
✓ 7	22:23:35	TRUNCATE TABLE customer_transactions	0 row(s) affected
✓ 8	22:24:10	CREATE TABLE bank_backup(txn_id INT, customer_name VARCHAR(100))	0 row(s) affected
✓ 9	22:24:46	DROP TABLE bank_backup	0 row(s) affected

Step 3: DML Commands

SQL File 4* x SQL File 4

```

27     txn_id INT,
28     customer_name VARCHAR(100)
29 );
30 -- Step 8: Drop Backup Table
31 • DROP TABLE bank_backup;
32 -- Step 9: Insert Sample Data
33 • INSERT INTO customer_transactions
34   (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
35   VALUES
36   (101, 'Angela', 'Kerala', 'Deposit', 5000, '2024-01-05'),
37   (102, 'Smija', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
38   (103, 'Andrea', 'Kerala', 'Deposit', 12000, '2024-01-08'),
39   (104, 'Priya', 'Vizag', 'Deposit', 8000, '2024-01-10'),
40   (105, 'Tianne', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
41   (106, 'Evelyn', 'Vizag', 'Deposit', 15000, '2024-01-12'),
42   (107, 'Advika', 'Kerala', 'Withdrawal', 1000, '2024-01-13'),
43   (108, 'Elizabeth', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
44   (109, 'George', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
45   (110, 'Mathew', 'Kerala', 'Deposit', 11000, '2024-01-16');
46

```

Output

Action Output

#	Time	Action	Message
✓ 5	12:30:44	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 6	12:30:49	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected
✓ 7	12:30:53	TRUNCATE TABLE customer_transactions	0 row(s) affected
✓ 8	12:30:59	CREATE TABLE bank_backup(txn_id INT, customer_name VARCHAR(100))	0 row(s) affected
✓ 9	12:31:02	DROP TABLE bank_backup	0 row(s) affected
✓ 10	12:38:41	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0


```

32 -- Step 9: Insert Sample Data
33 • INSERT INTO customer_transactions
34 (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
35 VALUES
36 (101, 'Angela', 'Kerala', 'Deposit', 5000, '2024-01-05'),
37 (102, 'Smija', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
38 (103, 'Andrea', 'Kerala', 'Deposit', 12000, '2024-01-08'),
39 (104, 'Priya', 'Vizag', 'Deposit', 8000, '2024-01-10'),
40 (105, 'Tianne', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
41 (106, 'Evelyn', 'Vizag', 'Deposit', 15000, '2024-01-12'),
42 (107, 'Advika', 'Kerala', 'Withdrawal', 1000, '2024-01-13'),
43 (108, 'Elizabeth', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
44 (109, 'George', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
45 (110, 'Mathew', 'Kerala', 'Deposit', 11000, '2024-01-16');
46 -- Step 10: Insert New Transaction
47 • INSERT INTO customer_transactions
48 (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
49 VALUES
50 (111, 'Venu', 'Vizag', 'Deposit', 7000, '2024-01-18');
51

```

Output				
Action Output				
#	Time	Action	Message	
✓ 6	12:30:49	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected	
✓ 7	12:30:53	TRUNCATE TABLE customer_transactions	0 row(s) affected	
✓ 8	12:30:59	CREATE TABLE bank_backup(txn_id INT, customer_name VARCHAR(100))	0 row(s) affected	
✓ 9	12:31:02	DROP TABLE bank_backup	0 row(s) affected	
✓ 10	12:38:41	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transa...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	
✓ 11	12:40:00	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transa...	1 row(s) affected	

Commands				
SQL File 4				
Limit to 1000 rows				
37	(102, 'Smija', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),			
38	(103, 'Andrea', 'Kerala', 'Deposit', 12000, '2024-01-08'),			
39	(104, 'Priya', 'Vizag', 'Deposit', 8000, '2024-01-10'),			
40	(105, 'Tianne', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),			
41	(106, 'Evelyn', 'Vizag', 'Deposit', 15000, '2024-01-12'),			
42	(107, 'Advika', 'Kerala', 'Withdrawal', 1000, '2024-01-13'),			
43	(108, 'Elizabeth', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),			
44	(109, 'George', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),			
45	(110, 'Mathew', 'Kerala', 'Deposit', 11000, '2024-01-16');			
46	-- Step 10: Insert New Transaction			
47	• INSERT INTO customer_transactions			
48	(txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)			
49	VALUES			
50	(111, 'Venu', 'Vizag', 'Deposit', 7000, '2024-01-18');			
51	-- Step 11: Update Transaction Amount			
52	• UPDATE customer_transactions			
53	SET amount = 5000			
54	WHERE txn_id = 105;			
55				
56				

Output				
Action Output				
#	Time	Action	Message	
✓ 7	12:30:53	TRUNCATE TABLE customer_transactions	0 row(s) affected	
✓ 8	12:30:59	CREATE TABLE bank_backup(txn_id INT, customer_name VARCHAR(100))	0 row(s) affected	
✓ 9	12:31:02	DROP TABLE bank_backup	0 row(s) affected	
✓ 10	12:38:41	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transa...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	
✓ 11	12:40:00	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transa...	1 row(s) affected	
✓ 12	12:41:09	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings:	

```
Commands* x SQL File 4
Limit to 1000 rows

39 (104,'Priya','Vizag','Deposit',8000,'2024-01-10'),
40 (105,'Tianne','Hyderabad','Withdrawal',3500,'2024-01-11'),
41 (106,'Evelyn','Vizag','Deposit',15000,'2024-01-12'),
42 (107,'Advika','Kerala','Withdrawal',1000,'2024-01-13'),
43 (108,'Elizabeth','Hyderabad','Deposit',9000,'2024-01-14'),
44 (109,'George','Vizag','Withdrawal',4000,'2024-01-15'),
45 (110,'Mathew','Kerala','Deposit',11000,'2024-01-16');
46 -- Step 10: Insert New Transaction
47 • INSERT INTO customer_transactions
48 (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
49 VALUES
50 (111,'Venu','Vizag','Deposit',7000,'2024-01-18');
51 -- Step 11: Update Transaction Amount
52 • UPDATE customer_transactions
53 SET amount = 5000
54 WHERE txn_id = 105;
55 -- Step 12: Delete Transaction
56 • DELETE FROM customer_transactions
57 WHERE txn_id = 111;
58
59
```

Output

Action Output

#	Time	Action	Message
✓ 8	12:30:59	CREATE TABLE bank_backup(txn_id INT, customer_name VARCHAR(100))	0 row(s) affected
✓ 9	12:31:02	DROP TABLE bank_backup	0 row(s) affected
✓ 10	12:38:41	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓ 11	12:40:00	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)	1 row(s) affected
✓ 12	12:41:09	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 13	12:42:24	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected

```
Commands* x SQL File 4
Limit to 1000 rows

52 • UPDATE customer_transactions
53 SET amount = 5000
54 WHERE txn_id = 105;
55 -- Step 12: Delete Transaction
56 • DELETE FROM customer_transactions
57 WHERE txn_id = 111;
58 -- Step 13: Display All Transactions
59 • SELECT *
60 FROM customer_transactions;
```

Result Grid

Filter Rows:

Edit: Export/Import: Wrap Cell Content:

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
▶	101	Angela	Kerala	Deposit	5000.00	2024-01-05	NULL
	102	Smija	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
	103	Andrea	Kerala	Deposit	12000.00	2024-01-08	NULL
	104	Priya	Vizag	Deposit	8000.00	2024-01-10	NULL
	105	Tianne	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL
	106	Evelyn	Vizag	Deposit	15000.00	2024-01-12	NULL
	107	Advika	Kerala	Withdrawal	1000.00	2024-01-13	NULL
	108	Elizabeth	Hyderabad	Deposit	9000.00	2024-01-14	NULL
	109	George	Vizag	Withdrawal	4000.00	2024-01-15	NULL
	110	Mathew	Kerala	Deposit	11000.00	2024-01-16	NULL

transactions 1 x

Output

Action Output

#	Time	Action	Message
✓ 9	12:31:02	DROP TABLE bank_backup	0 row(s) affected
✓ 10	12:38:41	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓ 11	12:40:00	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)	1 row(s) affected
✓ 12	12:41:09	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 13	12:42:24	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected
✓ 14	12:43:34	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned

Commands | SQL File 4

```

56 • DELETE FROM customer_transactions
57 WHERE txn_id = 111;
58 -- Step 13: Display All Transactions
59 • SELECT *
60 FROM customer_transactions;
61 -- Step 14: Display Deposit Transactions
62 • SELECT *
63 FROM customer_transactions
64 WHERE transaction_type = 'Deposit';

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: I A

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Angela	Kerala	Deposit	5000.00	2024-01-05	NULL	
103	Andrea	Kerala	Deposit	12000.00	2024-01-08	NULL	
104	Priya	Vizag	Deposit	8000.00	2024-01-10	NULL	
106	Evelyn	Vizag	Deposit	15000.00	2024-01-12	NULL	
108	Elizabeth	Hyderabad	Deposit	9000.00	2024-01-14	NULL	
110	Mathew	Kerala	Deposit	11000.00	2024-01-16	NULL	
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL

customer_transactions 2 x Apply

Output

Action Output

#	Time	Action	Message
✓ 10	12:38:41	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, trans...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓ 11	12:40:00	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, trans...	1 row(s) affected
✓ 12	12:41:09	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 13	12:42:24	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected
✓ 14	12:43:34	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned
✓ 15	12:44:41	SELECT * FROM customer_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000	6 row(s) returned

Commands | SQL File 4

```

58 -- Step 13: Display All Transactions
59 • SELECT *
60 FROM customer_transactions;
61 -- Step 14: Display Deposit Transactions
62 • SELECT *
63 FROM customer_transactions
64 WHERE transaction_type = 'Deposit';
65 -- Step 15: Sort Transactions by Amount
66 • SELECT *
67 FROM customer_transactions
68 ORDER BY amount DESC;
69 -- Create Auditor1 User
70 • CREATE USER 'Auditor1'@'localhost'
71 IDENTIFIED BY '1234';
72
73 -- Create BranchManager User
74 • CREATE USER 'BranchManager'@'localhost'
75 IDENTIFIED BY '1234';
76
77

```

Output

Action Output

#	Time	Action	Message
✓ 11	12:40:00	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, trans...	1 row(s) affected
✓ 12	12:41:09	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 13	12:42:24	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected
✓ 14	12:43:34	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned
✓ 15	12:44:41	SELECT * FROM customer_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000	6 row(s) returned
✓ 16	12:45:26	CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected

Step 4: DCL Commands

The screenshot displays two screenshots of the SQL Developer interface, showing the execution of DCL (Data Control Language) commands. The top screenshot shows the first 21 lines of the script, and the bottom screenshot shows the next 7 lines (lines 17-22).

Top Screenshot:

```
61 -- Step 14: Display Deposit Transactions
62 • SELECT *
63 FROM customer_transactions
64 WHERE transaction_type = 'Deposit';
65 -- Step 15: Sort Transactions by Amount
66 • SELECT *
67 FROM customer_transactions
68 ORDER BY amount DESC;
69 -- Create Auditor1 User
70 • CREATE USER 'Auditor1'@'localhost'
71 IDENTIFIED BY '1234';
72
73 -- Create BranchManager User
74 • CREATE USER 'BranchManager'@'localhost'
75 IDENTIFIED BY '1234';
76 -- Step 16: Grant SELECT Permission
77 • GRANT SELECT
78 ON customer_transactions
79 TO 'Auditor1'@'localhost';
80
81
```

Output:

#	Time	Action	Message
✓ 16	12:45:26	CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected
✗ 17	13:53:58	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT
✗ 18	13:54:38	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT
✓ 19	13:54:57	CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected
✗ 20	13:54:59	CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY '1234'	Error Code: 1396. Operation CREATE USER failed for 'BranchManager'@'localhost'
✓ 21	13:55:02	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	0 row(s) affected

Bottom Screenshot:

```
65 -- Step 15: Sort Transactions by Amount
66 • SELECT *
67 FROM customer_transactions
68 ORDER BY amount DESC;
69 -- Create Auditor1 User
70 • CREATE USER 'Auditor1'@'localhost'
71 IDENTIFIED BY '1234';
72
73 -- Create BranchManager User
74 • CREATE USER 'BranchManager'@'localhost'
75 IDENTIFIED BY '1234';
76 -- Step 16: Grant SELECT Permission
77 • GRANT SELECT
78 ON customer_transactions
79 TO 'Auditor1'@'localhost';
80 -- Step 17: Grant All Privileges
81 • GRANT ALL
82 ON customer_transactions
83 TO 'BranchManager'@'localhost';
84
85
```

Output:

#	Time	Action	Message
✗ 17	13:53:58	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT
✗ 18	13:54:38	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT
✓ 19	13:54:57	CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected
✗ 20	13:54:59	CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY '1234'	Error Code: 1396. Operation CREATE USER failed for 'BranchManager'@'localhost'
✓ 21	13:55:02	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	0 row(s) affected
✓ 22	13:56:04	GRANT ALL ON customer_transactions TO 'BranchManager'@'localhost'	0 row(s) affected

Commands* SQL File 4

Limit to 1000 rows

```

69 -- Create Auditor1 User
70 • CREATE USER 'Auditor1'@'localhost'
71 IDENTIFIED BY '1234';
72
73 -- Create BranchManager User
74 • CREATE USER 'BranchManager'@'localhost'
75 IDENTIFIED BY '1234';
76 -- Step 16: Grant SELECT Permission
77 • GRANT SELECT
78 ON customer_transactions
79 TO 'Auditor1'@'localhost';
80 -- Step 17: Grant All Privileges
81 • GRANT ALL
82 ON customer_transactions
83 TO 'BranchManager'@'localhost';
84 -- Step 18: Revoke SELECT Permission
85 • REVOKE SELECT
86 ON customer_transactions
87 FROM 'Auditor1'@'localhost';
88
89

```

Output

Action Output

#	Time	Action	Message
✗ 18	13:54:38	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT
✓ 19	13:54:57	CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected
✗ 20	13:54:59	CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY '1234'	Error Code: 1396. Operation CREATE USER failed for 'BranchManager'@'localhost'
✓ 21	13:55:02	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	0 row(s) affected
✓ 22	13:56:04	GRANT ALL ON customer_transactions TO 'BranchManager'@'localhost'	0 row(s) affected
✓ 23	13:56:46	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@'localhost'	0 row(s) affected

Commands* SQL File 4

Limit to 1000 rows

```

73 -- Create BranchManager User
74 • CREATE USER 'BranchManager'@'localhost'
75 IDENTIFIED BY '1234';
76 -- Step 16: Grant SELECT Permission
77 • GRANT SELECT
78 ON customer_transactions
79 TO 'Auditor1'@'localhost';
80 -- Step 17: Grant All Privileges
81 • GRANT ALL
82 ON customer_transactions
83 TO 'BranchManager'@'localhost';
84 -- Step 18: Revoke SELECT Permission
85 • REVOKE SELECT
86 ON customer_transactions
87 FROM 'Auditor1'@'localhost';
88 -- Step 19: Revoke All Privileges
89 • REVOKE ALL PRIVILEGES
90 ON customer_transactions
91 FROM 'BranchManager'@'localhost';
92
93

```

Output

Action Output

#	Time	Action	Message
✓ 19	13:54:57	CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected
✗ 20	13:54:59	CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY '1234'	Error Code: 1396. Operation CREATE USER failed for 'BranchManager'@'localhost'
✓ 21	13:55:02	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	0 row(s) affected
✓ 22	13:56:04	GRANT ALL ON customer_transactions TO 'BranchManager'@'localhost'	0 row(s) affected
✓ 23	13:56:46	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@'localhost'	0 row(s) affected
✓ 24	13:58:12	REVOKE ALL PRIVILEGES ON customer_transactions FROM 'BranchManager'@'localhost'	0 row(s) affected

Step 5:TCL Commands

The screenshot shows the SQL Developer interface with a script titled "SQL File 4". The script contains the following SQL commands:

```
80 -- Step 17: Grant All Privileges
81 • GRANT ALL
82   ON customer_transactions
83   TO 'BranchManager'@'localhost';
84 -- Step 18: Revoke SELECT Permission
85 • REVOKE SELECT
86   ON customer_transactions
87   FROM 'Auditor1'@'localhost';
88 -- Step 19: Revoke All Privileges
89 • REVOKE ALL PRIVILEGES
90   ON customer_transactions
91   FROM 'BranchManager'@'localhost';
92 -- Step 20: Commit Transaction
93
94 • UPDATE customer_transactions
95   SET amount = 2000
96   WHERE txn_id = 102;
97
98 • COMMIT;
```

The output window shows the execution results for steps 20 through 25:

#	Time	Action	Message
20	13:54:59	CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY '1234'	Error Code: 1396. Operation CREATE USER failed for 'BranchManager'@'localhost'
21	13:55:02	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	0 row(s) affected
22	13:56:04	GRANT ALL ON customer_transactions TO 'BranchManager'@'localhost'	0 row(s) affected
23	13:56:46	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@'localhost'	0 row(s) affected
24	13:58:12	REVOKE ALL PRIVILEGES ON customer_transactions FROM 'BranchManager'@'localhost'	0 row(s) affected
25	13:59:18	COMMIT	0 row(s) affected

The screenshot shows the SQL Developer interface with a script titled "SQL File 4". The script contains the following SQL commands:

```
84 -- Step 18: Revoke SELECT Permission
85 • REVOKE SELECT
86   ON customer_transactions
87   FROM 'Auditor1'@'localhost';
88 -- Step 19: Revoke All Privileges
89 • REVOKE ALL PRIVILEGES
90   ON customer_transactions
91   FROM 'BranchManager'@'localhost';
92 -- Step 20: Commit Transaction
93
94 • UPDATE customer_transactions
95   SET amount = 2000
96   WHERE txn_id = 102;
97
98 • COMMIT;
99 -- Step 21: Create Savepoint
100 • START TRANSACTION;
101
102 • SAVEPOINT Before_Update;
```

The output window shows the execution results for steps 21 through 26:

#	Time	Action	Message
21	13:55:02	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	0 row(s) affected
22	13:56:04	GRANT ALL ON customer_transactions TO 'BranchManager'@'localhost'	0 row(s) affected
23	13:56:46	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@'localhost'	0 row(s) affected
24	13:58:12	REVOKE ALL PRIVILEGES ON customer_transactions FROM 'BranchManager'@'localhost'	0 row(s) affected
25	13:59:18	COMMIT	0 row(s) affected
26	13:59:53	SAVEPOINT Before_Update	0 row(s) affected

Commands* x SQL File 4

Limit to 1000 rows

```

93
94 • UPDATE customer_transactions
95   SET amount = 2000
96   WHERE txn_id = 102;
97
98 • COMMIT;
99   -- Step 21: Create Savepoint
100 • START TRANSACTION;
101
102 • SAVEPOINT Before_Update;
103   -- Step 22: Rollback to Savepoint
104 • UPDATE customer_transactions
105   SET amount = 99999
106   WHERE txn_id = 102;
107
108 • ROLLBACK TO SAVEPOINT Before_Update;
109   -- Reset Transaction State
110 • ROLLBACK;
111
112 • SET autocommit = 1;
113

```

Output

Action Output

#	Time	Action	Message
22	13:56:04	GRANT ALL ON customer_transactions TO 'BranchManager'@'localhost'	0 row(s) affected
23	13:56:46	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@'localhost'	0 row(s) affected
24	13:58:12	REVOKE ALL PRIVILEGES ON customer_transactions FROM 'BranchManager'@'localhost'	0 row(s) affected
25	13:59:18	COMMIT	0 row(s) affected
26	13:59:53	SAVEPOINT Before_Update	0 row(s) affected
27	14:00:34	SET autocommit = 1	0 row(s) affected

Commands* x SQL File 4

Limit to 1000 rows

```

109   -- Reset Transaction State
110 • ROLLBACK;
111
112 • SET autocommit = 1;
113   -- Step 23: Complete TCL Example
114 • START TRANSACTION;
115
116 • UPDATE customer_transactions
117   SET amount = 7000
118   WHERE txn_id = 101;
119
120 • SAVEPOINT SP1;
121
122 • UPDATE customer_transactions
123   SET amount = 9000
124   WHERE txn_id = 102;
125
126 • ROLLBACK TO SAVEPOINT SP1;
127
128 • COMMIT;
129

```

Output

Action Output

#	Time	Action	Message
23	13:56:46	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@'localhost'	0 row(s) affected
24	13:58:12	REVOKE ALL PRIVILEGES ON customer_transactions FROM 'BranchManager'@'localhost'	0 row(s) affected
25	13:59:18	COMMIT	0 row(s) affected
26	13:59:53	SAVEPOINT Before_Update	0 row(s) affected
27	14:00:34	SET autocommit = 1	0 row(s) affected
28	14:01:16	COMMIT	0 row(s) affected

